HILA SABOUNI

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HCI PhD Candidate | Mixed-Methods Research | Human-AI Interaction | Product & Behavioral Research

HCI researcher with academic and industry experience studying how people interpret, trust, and make decisions with emerging technologies. Experienced in experimental design, interviews, surveys, usability testing, qualitative and quantitative analysis, and translating behavioral insights into actionable product recommendations.

EDUCATION

Ph.D. in Human-Computer Interaction (HCI) | *Iowa State University, Ames, IA* — *2023–May 2027 (Expected)*

M.S. in Human-Computer Interaction (HCI) | *Iowa State University* — *2022–2023*

Master of Architecture (M.Arch.) | *Iowa State University* — *2020–2022*

EXPERIENCE

CUSTOMER EXPERIENCE ANALYST INTERN | ROCKET MORTGAGE *|Detroit, MI* — *May 2025–Aug 2025*

- Built AI-driven qualitative tagging frameworks using prompt engineering to accelerate analysis of large-scale customer feedback.
- Analyzed 2023–2025 NPS data using qualitative and sentiment analysis, identifying three key behavioral drivers influencing product and marketing decisions.
- Synthesized customer motivations and pain points into five actionable product recommendations for improving the digital mortgage journey.
- Won Best Use of AI for Discovery & Research in a company-wide hackathon with 10+ competing teams.

HCI RESEARCHER & DESIGNER *|Iowa State University* — *Aug 2022–Present*

SOCIAL INFLUENCE & DECISION-MAKING WITH AI

- Designed and led multi-phase mixed-method studies with 150+ participants examining how people interpret, trust, and make decisions with AI-generated outputs.
- Developed controlled experiments to investigate how social and psychological cues influence user confidence, perception, and reliance on AI.
- Conduct interviews, surveys, and usability-style evaluations, applying thematic and content analysis to identify behavioral patterns and user attitudes.
- Translate findings into research-based recommendations and frameworks that support effective human-AI interaction, decision-making, and cross-functional collaboration.

COWXR: VR TRAINING FOR HIGH-STAKES DECISIONS

- Evaluated VR-based training for high-stakes veterinary procedures through a between-subjects study comparing immersive VR training with instructor-led training.
- Examined differences in learner confidence and task performance to assess how immersive training affects preparedness for high-pressure decision-making.
- Found that participants trained in VR reported greater confidence and achieved higher performance scores, supporting VR as a complement to traditional training.

SENSORY CUES IN VIRTUAL REALITY

- Designed and led a 71-participant mixed-method study comparing haptic, audio, visual, and control conditions in a VR visual-search task.
- Built the experimental environment in Rhino and Unity and conducted participant training, testing, and surveys.
- Analyzed quantitative and qualitative data to evaluate performance, user experience, and distraction across sensory conditions.
- Found haptic guidance performed best without adding visual distraction, with audio ranking second.

XR GAME DESIGN FOR CYBERSECURITY LEARNING

- Contributed to brainstorming, narrative development, and environment design for an XR cybersecurity game.
- Conducted interviews and user testing to identify learner needs and refine the experience for rural English learners.
- Collaborated across HCI, education, and cybersecurity to integrate research insights into the game design.

RESEARCH & 3D MODELING MENTOR | NSF REU *Summer 2023 & 2024*

- Mentored research teams in 3D modeling, game development, and user-centered design for STEM education.
- Guided interviews, focus groups, surveys, and prototype testing to identify user needs and learning behaviors.
- Taught Blender and supported the development of interactive 3D learning experiences and a mini-game.
- Helped teams translate research findings into clearer design, storytelling, and learning strategies.

ADDITIONAL EXPERIENCE

UNIVERSITY INSTRUCTOR | ARCHITECTURE STUDIO COURSE | *Ames, IA* *Aug 2022–May 2024*

- Led studio instruction, critiques, and workshops focused on user-centered design, problem-solving, and design communication.

ARCHITECT INTERN | KLINGNER & ASSOCIATES, PC | *Pella, IA* *Jan 2022–Aug 2022*

- Collaborated with clients and cross-functional teams and produced technical drawings, 3D models, and documentation supporting project delivery.

FOUNDER, OWNER, AND DESIGNER | RISEN DESIGN | *Iran* *May 2016–Nov 2018*

- Led end-to-end design and delivery of nine industrial and residential projects, achieving up to 10% cost savings and 20–30% fewer revisions.
- Received the City Diploma of Customer Rights Protection for excellence in client satisfaction and service quality.

SELECTED PUBLICATIONS

- **VR Guidance Cues in Virtual Environments –** *Journal of Perceptual Imaging*, 2025
- **Apple's Knowledge Navigator: Why Doesn't That Conversational Agent Exist Yet? –** *ACM CHI, 2024* (Honorable Mention)
- **Human Perception of Floor Plan Evacuatability –** *HFES Annual Meeting*, 2024

KEY SKILLS

- **Research Methods:** Mixed-Methods Research, Experimental Design, Interviews, Surveys, Usability Testing, Contextual Inquiry, Task Analysis, Focus Groups
- **Analysis:** Thematic Analysis, Content Analysis, Sentiment Analysis, Regression, Correlation, t-tests, ANOVA
- **AI & Research:** Prompt Engineering, AI-Assisted Qualitative Analysis, Human-AI Interaction Research
- **Tools:** Qualtrics, SPSS, Taguette, Unity, Rhino 3D, Blender, Figma, Miro, Adobe Creative Suite
- **Languages:** Persian, English, Turkish

LEADERSHIP & AWARDS

President & Vice President | HCI Student Organization, Iowa State University *2023– 2025*

- Expanded professional and social programming, tripling community engagement and reviving the organization's flagship Usabilathon competition.

Best Use of AI for Discovery & Research – *Hackathon at Rocket Mortgage, 2025*
Honorable Mention – *ACM CHI Conference, 2024*
City Diploma of Customer Right Protection – *Risen Design, 2018*